Dazhong Shen

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RESEARCH INTERESTS

My general research interest includes data mining and machine learning, focusing on generative models, variational inference, graph neural networks, and their data intensive applications.

EDUCATION

Ph.D. Sep. 2017 $\sim$ *Expected:* June 2023
Computer Science, University of Science and Technology of China (USTC)
Advisor: Prof. Hui Xiong, AAAS/IEEE Fellow
Major: Computer Science in Application

B.S. Sep. 2013 $\sim$ Jul. 2017
School of Mathematical Sciences, University of Science and Technology of China (USTC)
Major: Computational Mathematics, GPA: 3.63/4.30 (Major 3.82/4.30)

PROFESSIONAL EXPERIENCE

Research Intern, Talent Intelligence Center, Baidu Inc.
Mar. 2017 $\sim$ Sep. 2017 and Aug. 2018 $\sim$ June 2022, Mentor: Dr. Hengshu Zhu

- Developed a novel neural topic model with Graph Neural Networks by transforming each document into a directed graph with word dependency as edges, where **a well-defined generative story for the document graph is designed**, and **the topic can be visualized in a graph manner**. The related paper has been published on NeurIPS-2021.
- Proposed a simple but effective and generalized strategy to avoid the posterior collapse in variational autoencoder by collecting the geometric properties of latent space, i.e., diversity and uncertainty, where **Batch Normalization and Dropout are applied on the posterior's parameters**. The related paper has been published on IJCAI-2021.
- Developed two topic models (traditional and neural) to jointly represent multiple textual collects, where both hierarchical relationships among them and individual characteristics were modeled in a probabilistic manner. **The algorithm is launched in Baidu campus recruitment system to enhance talent searching, person-job fitting, and interview question recommendation**. Related papers have been published on IJCAI-2018 and ACM TOIS.
- Designed a novel Graph Convolution Nerual with the maximization operation to **handle the over-smoothing and over-squashing problems** simultaneously. The related paper was submitted to NeurIPS-2022

Data Contest, Aftershock Detection Contest, Directed by China Earthquake Administration
May 2017 $\sim$ Oct. 2017

- Developed an ensemble learning-based framework for identifying seismic P-phase arrivals automatically of the aftershocks. Based on our great performance, we won **the championship in the First Season and the runner-up in the Finals from 1,143 teams participated from around the world**. The related work was also published on SCIENCE CHINA Information Sciences (Chinese).

Research Assistant, The Hong Kong University of Science and Technology (Guangzhou)
Oct. 2022 $\sim$ Present, Advisor: Prof. Hui Xiong

- Plan to do some interesting research works about graph neural network, generative models, or causal inference.

AWARDS

- China National Scholarship. (for Ph.D. students) 2022
- Huawei Scholarship (USTC). (for postgraduate and Ph.D. students) 2018, 2021
- Global Digital Technology Scholarship. 2018
- IJCAI-2018 Student Travel Grants. 2018
- Aftershock Detection Contest (China Earthquake Administration), Runner-Up. 2017
- Academic Scholarship (USTC), Grad 1. six times from 2017 to 2022
- Zhang Zongzhi Science and Technology Scholarship (USTC). 2016
- Outstanding Student Scholarship (USTC), Silver Medal. 2015
- Outstanding Student Scholarship (USTC), Bronze Medal. 2014

PUBLICATIONS (7 CCF A, 1 CCF A Chinese Journal)

1. **Dazhong Shen**, Chuan Qin, Hengshu Zhu, Tong Xu, Enhong Chen, Hui Xiong. Joint Representation Learning with Relation-enhanced Topic Models for Intelligent Job Interview Assessment, ACM Transactions on Information Systems **(ACM TOIS)**, 2021. **(CCF A)**
2. **Dazhong Shen**, Chuan Qin, Chao Wang, Zheng Dong, Hengshu Zhu, Hui Xiong. Topic Modeling Revisited: A Document Graph-based Neural Network Perspective. The 35th Conference on Neural Information Processing Systems **(NeurIPS-2021)**, 2021. **(CCF A)**
3. **Dazhong Shen**, Chuan Qin, Chao Wang, Hengshu Zhu, Enhong Chen, Hui Xiong. Regularizing Variational Autoencoder with Diversity and Uncertainty Awareness. The 30th International Joint Conference on Artificial Intelligence **(IJCAI-2021)**, 2021. **(CCF A)**
4. **Dazhong Shen**, Hengshu Zhu, Chen Zhu, Tong Xu, Chao Ma, Hui Xiong. A Joint Learning Approach to Intelligent Job Interview Assessment, The 27th International Joint Conference on Artificial Intelligence **(IJCAI-2018)**, 2018. **(CCF A)**
5. 申大忠, 张琦, 徐童, 祝恒书, 赵雯佳, 殷子凯, 周培伦, 房立华, 陈恩红, 熊辉. EL-Picker：基于集成学习的余震P波初动实时拾取方法, 《中国科学: 信息科学》, 2020. **(CCF A Chinese)**
6. Qingxin Meng, Keli Xiao, **Dazhong Shen**, Hengshu Zhu, Hui Xiong, Fine-Grained Job Salary Benchmarking with a Nonparametric Dirichlet-process-based Latent Factor Model. INFORMS Journal on Computing **(JOC)**, 2022. **(UTD-24)**
7. Chuan Qin, Kaichun Yao, Hengshu Zhu, Tong Xu, **Dazhong Shen**, Enhong Chen, Hui Xiong. Towards Automatic Job Description Generation with Capability-Aware Neural Networks, IEEE Transactions on Knowledge and Data Engineering **(IEEE TKDE)**, 2022. **(CCF A)**
8. Zhi Zheng, Chao Wang, Tong Xu, **Dazhong Shen**, Penggang Qin, Xiangyu Zhao, Baoxing Huai, Xian Wu, Enhong Chen. Interaction-aware Drug Package Recommendation via Policy Gradient. ACM Transactions on Information Systems **(ACM TOIS)**, 2022. **(CCF A)**

9. Zhi Zheng, Chao Wang, Tong Xu, **Dazhong Shen**, Penggang Qin, Baoxing Huai, Tongzhu Liu, Enhong Chen. Drug Package Recommendation via Interaction-aware Graph Induction. The 30th International World Wide Web Conference **(WWW-2021)**, 2021. **(CCF A)**

Under Review:

10. **Dazhong Shen**, et al. Handling Over-Smoothing and Over-Squashing in Graph Convolution with Maximization Operation. IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Submitted.
11. **Dazhong Shen**, et al. Exploiting Connections among Personality, Job Position, and Work Behavior: Evidence from Joint Bayesian Learning. ACM Transactions on Management Information Systems (TMIS), Submitted.

PATENTS

1. 申大忠，祝恒书，马超，朱琛，熊辉. 基于人工智能的辅助面试方法、装置及存储介质：中国，CN110472647[A]. 2019-11-19.
2. 申大忠，秦川，王超，董政，祝恒书，熊辉. 文本主题的确定方法、装置及电子设备: 中国，CN114492393[A]. 2022-05-13.
3. 秦川，姚开春，祝恒书，马超，申大忠. 一种招聘岗位描述文本的生成方法、装置、设备及介质: 中国，CN113627135[A]. 2021-11-09.
4. 苏昱涵，秦川，申大忠，赵洪科，宋欣，祝恒书. 用于处理技能信息的方法、模型训练方法及装置: 中国，CN114330333[A]. 2022-04-12.
5. 查睿，秦川，张乐，申大忠，徐童，祝恒书. 职业流动关系的预测方法、系统、设备及存储介质: 中国，CN114331380[A]. 2022-04-12
6. Chuan Qin, Kaichun Yao, Hengshu Zhu, Chao Ma, **Dazhong Shen**. Method, apparatus, device and medium for generating recruitment position description text: US, US20210216726[A]. 2021-07-15.

ACADEMIC SERVICES

Program Committees Member:

- The 39th International Conference on Machine Learning (ICML-2022)
- The 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-2022)
- The 36th Conference on Neural Information Processing Systems (NeurIPS-2022)
- The 37th AAAI Conference on Artificial Intelligence (AAAI-2023)

Sub-reviewer for Journals and Conferences:

- IEEE TKDE, ACM TOIS, IEEE TSMC, etc.
- ICML, NeurIPS, KDD, WWW, SIGIR, AAAI, IJCAI, CVPR, ICDM, CIKM, WSDM, etc.